



CRISPR detection in metagenome-assembled genomes (MAGs) of coal mine

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Abstract

Bacterial and archaeal CRISPR-Cas systems provide adaptive immune protection against foreign mobile genetic elements. When viruses infect bacteria, a small portion of the viral DNA is inserted into the bacterial DNA in a specific pattern to produce segments known as CRISPR arrays. Metagenome assembled genomes (MAGs) were used in our study to identify the CRISPR sequence for determining the interacted phage. Metagenomic data from a coal mine was used to perform a computational study. From raw reads, 206151 contigs were assembled. Then contigs were clustered into 150 Metagenome assembled genomes from which 78 non-redundant MAGs were selected. Using the CHECKM standard, seven MAGs were found to have >80 completeness and <20 contaminations. Those MAGs were analyzed for the presence of CRISPR elements. Out of seven MAGs, four MAGs have the CRISPR elements and are searched against the VIROblast database. CRISPR arrays have 4, 1, 3, and 7 spacer sequences in the MAGs of *Burkholderia*, *Acinetobacter*, *Oxalobacteraceae*, and *Burkholderia multivorans* respectively. The uncultured *Caudovirales* phage genomic regions were present in the genomes of *Burkholderia*, *Oxalobacteriaceae*, and *Burkholderia multivorans*. This study follows the unconventional metagenomics workflow to provide a better understanding of bacteria and phage interactions.

Keywords Metagenomics · CRISPR sequence · Bacteriophage · Metagenome assembled genomes (MAGs) · Spacer sequences

Introduction

Bacteriophages are the most prevalent and varied organisms on the planet; they are contagious viruses that replicate only in bacterial hosts. A bacteriophage's life cycle completely depends on the bacteria it infects (Rohwer 2003). Bacteriophages are usually found in soil, oceans, deserts, wastewater, sewage, the guts of animals, and activated sludge (Koskella and Meaden 2013). Most microbial communities' stability is due to the interaction between phages and their hosts (Hayes et al. 2017). It plays a major role in host mortality, carbon cycling, nutrient cycling, horizontal gene transfer, and structuring food webs by providing protists with nutrients (Stone et al. 2019).

The majority of archaea and 40% of sequenced bacterial species have clustered regularly interspaced short palindromic repeats (CRISPR) along with CRISPR-associated (Cas) genes, creating a defence system against bacteriophage infection (Huo et al. 2023; Najjar et al. 2020). As part of their defence mechanism, CRISPR loci are made up of small palindromic repeating units and several unique spacer sequences that accumulate over time due to infections by viruses (Barrangou et al. 2007). After phage infection, genomic fragments are integrated nearer to the leading end of the CRISPR array as spacers. (Karginov and Hannon 2010). The complementary nucleic acid sequences known as protospacers are transcribed from these spacer sequences and processed into CRISPR RNAs (crRNAs), which work with Cas genes like the DNA exonuclease Cas to recognise and cleave protospacers, providing microorganisms with an adaptable molecular immune system (Rath et al. 2015; De Sordi et al. 2019).

Detecting and recovering genomic sequences from uncultivated phage and linking them to their hosts is challenging in ecological investigations (Hatfull and Hendrix 2011;

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Thakur et al. 2023). The CRISPR locus can help with both problems. Population-level studies of CRISPR loci can reveal details about the dynamics of phage-host interactions (Zhang et al. 2021). Understanding phage populations in an ecosystem comprises the experimental methodology, i.e. sampling and DNA extraction, which is not feasible in vitro as it requires a host for the whole process and it is challenging with its pathogenicity. Our workflow aims to utilise existing bacterial metagenomic data to get insights into the phage diversity by computational approach.

Materials and methods

DNA isolation and shotgun sequencing

The lignite sample was collected from the coal mine in Neyveli, Tamil Nadu (11° 35' 34.44" N and 79° 29' 29.04" E). The DNeasy PowerMax Soil extraction kit was used to extract metagenomic DNA from the sample. The Metagenome library was prepared and sequenced on Illumina Novoseq 6000 platform paired-end sequencing (2 × 150 bp) from CytoGene® Shotgun metagenome sequencing services

(Research & Development LLP, Lucknow) and the raw reads were generated in fastq files.

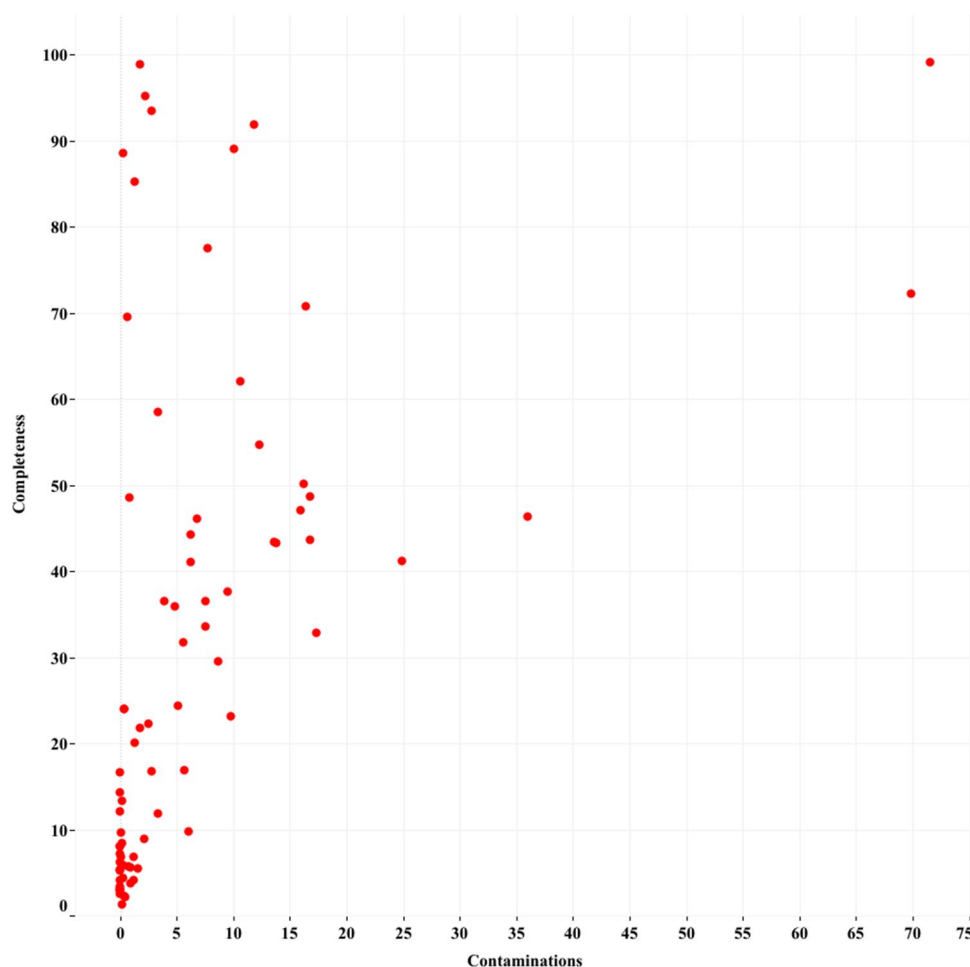
Quality analysis

The following steps were taken to improve the quality of the raw reads. The first step was to use FastQC version 0.11.9 (Andrews 2010) to check the quality of the raw reads (.fastq) and MultiQC version 1.8 (Ewels et al. 2016). The poor-quality reads with a Phred score lower than Q30, reads less than 50 bp in size, and adapter content were trimmed using Trimmomatic version 0.39 (Bolger et al. 2014) with the parameters of PE, -phred33, ILLUMINACLIP:adapter.fa:2:30:10, LEADING:3, TRAILING:3, SLIDINGWINDOW:4:15, and MINLEN:50 with the paired-end (forward and reverse) fastq files.

Pre-processing of the raw reads

Subsequently, the clean reads were used as input for the MEGAHIT v1.2.9 assembler (Li et al. 2015). The fastq reads co-assembly was carried out using the options -1, -2, -k-list 21, 29, 39, 59, 79, 99, 119, 141, -min-contig-len 500, -o

Fig. 1 Red dots represent each bin to validate the completeness and contamination of MAGs



assembly.fa, and -t 15. The assembled contigs file (.fasta) was used as input in QUAST version 5.2.0 (Gurevich et al. 2013) to check the numbers and length of the contigs.

Genomes' reconstruction

Metagenomic binning methods were followed to group the contigs to reconstruct a metagenome-assembled genome (MAG). For creating and evaluating MAGs, SqueezeM-eta (SQM) pipeline v1.6.0 (Tamames and Puente-Sánchez 2019) was followed, and it used binning tools such as MetaBAT and MaxBin and contigs file (.fasta) as input. With the parameter of -m coassembly, -p MAGs, -s samplefile, -f files.fastq, -extassembly file.fasta, -binner maxbin, metabat, and -t 16, it generates metagenome-assembled genomes (MAGs), called bins (.fasta), as output. Binner tools such as MaxBin (Wu et al. 2014) use the expectation maximisation algorithm to

group contigs, whereas MetaBAT (Kang et al. 2019) uses the label propagation algorithm. Contigs were calculated in tetranucleotide frequencies and reconstructed into genomes of similar tetranucleotide frequencies.

Bin's quality

Bins (.fasta) were used as input in DAS tool (Sieber et al. 2018) to filter out the redundant bins from the assembly. The generated output was non-redundant bins in FASTA format. The resulting bins were assessed for completeness and contamination with CheckM (Parks et al. 2015) with the default options, and the output generated was tab separated value (.tsv) file containing the taxonomy labelled with bins and the values of completeness and contamination of a genome calculated in a percentage score. CheckM uses single-copy genes to evaluate a genome; i.e. a genome is 100%

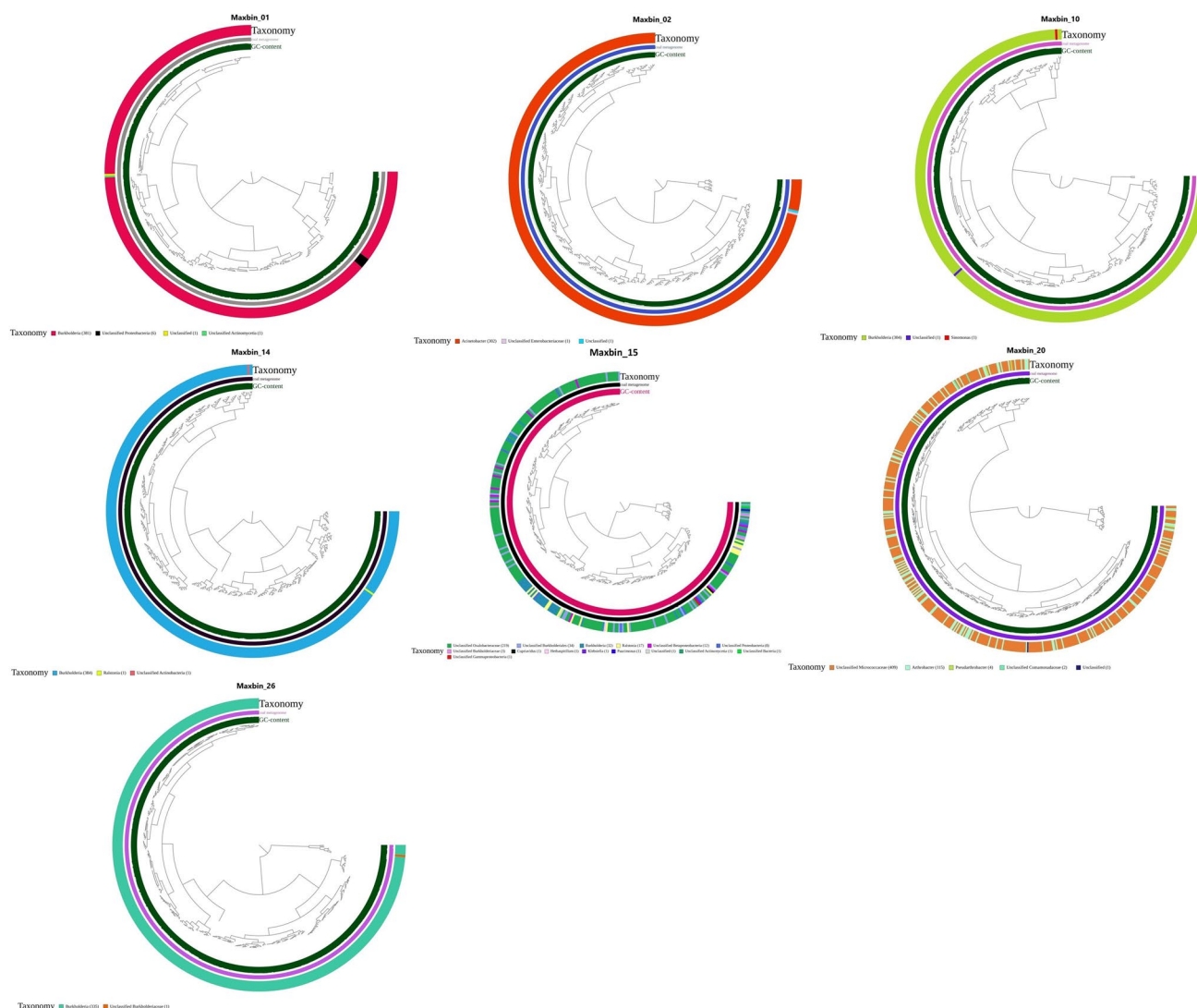


Fig. 2 Seven quality metagenome-assembled genomes (MAGs) with their contaminations

complete if all the genes are present, and it is contaminated if a gene appears more than once. The bins with completeness $\geq 90\%$ and contamination $\leq 10\%$ were called golden bins (Regan et al. 2018). Then the SQM results were loaded into Anvi'o (Eren et al. 2015) with the option `anvi-load-sqm.py, -p MAGs, -o anvio, -num-threads 12, -min-contig-length 2500, -run-scg-taxonomy, -run-HMMS`, to create CONTIGS.db, and PROFILE.db. These databases contain information on contigs, K-mer frequencies, annotated genes, and sample information. Using the option, `anvi-refine, -c CONTIGS.db, -p PROFILE.db, -C DAS, and -b Bins`, the selected quality bins were visualized in HTML format.

CRISPR element and phage detection

The selected quality bins (.fasta) were further inspected in the MinCED v0.4.2 tool (Skenner et al. 2019; Bland et al. 2007) for mining the CRISPR elements. It generates an output of repeat and spacer sequences in text file format. The spacer sequences were extracted from the MAGs and listed in a separate file (.fasta). The fasta files were uploaded in ViroBLAST server (Deng et al. 2007) to determine the phage interaction with the genomes. CRISPR-Cas systems i.e. classes, types, and subtypes were identified for quality bins using CRISPRone tool with the input of MAGs in the fasta file.

Results and discussion

Quality analysis

The Illumina platform produced 93,668,322 raw reads with a GC content of 60.59%. The average Phred score for paired-end reads was Q35, with a mean sequence length of 150 bp. The quality reads were assembled following quality filtering, which involved removing the adapter and trimming the low-base quality.

Pre-processing of the raw reads

A total of 206,151 contigs were assembled by the MEGA-HIT assembler. The generated contigs' were evaluated by the QUAST tool results in 226 contigs with $> 50,000$ bp, 1240 contigs with $> 25,000$ bp, 5265 with $> 10,000$ bp, 11,596 contigs with $> 5,000$ bp, 73,144 contigs with $> 1,000$ bp, and 114,680 contigs with > 500 bp.

Genomes' reconstruction and Bin's quality

Contigs were grouped into clusters to represent the whole genome of an organism using binning tools. MetaBAT and MaxBin produced 83 and 67 bins, respectively.

Out of 150 bins, 78 non-redundant bins were identified (Fig. 1) and assigned to their taxonomy. Then the bins were filtered with CheckM standard completeness $\geq 80\%$ and contamination $\leq 20\%$. Following the standards, 71 bins had low completeness or high contamination, while 7 bins adhered to this standard (Fig. 2). In brief, CheckM list showed that the standard metagenome-assembled genomes (MAGs) were from MaxBin and not from MetaBAT.

CRISPR array and phage detection

The minCED tool identified out of seven MAGs, only four MAGs—MaxBin 01,02,15, and 26 have CRISPR arrays. A total of 23 arrays were identified, of which MaxBin 01 possesses the highest number of arrays. On analysis of spacers in the individual array, five arrays have more than one spacer sequence. The number of arrays for each MaxBin with the spacer sequence is shown in Table 1. All 39 spacer sequences were extracted and used as queries in the ViroBLAST server to identify the interacting phage. Of the 39 spacers, 17 spacers were identified as viral sequences, and 22 spacer sequences did not show any hit. The spacer sequence and the hits are shown in Table 1. Uncultured *Caudovirales* phage was found to interact with the genomes of *Burkholderia*, *Oxalobacteriaceae*, and *Burkholderia multivorans*. *Caudovirales* phage is a double-stranded DNA phage that would have delivered the DNA by following the injectosome mechanism to the bacterial genomes. From the kingdom of *Heunggongvirae* and phylum of *Urovicota*, the *Caudovirales* has 4 orders, 47 families, 98 subfamilies, 1197 genera, and 3601 species in the class. This makes *Caudovirales* one of the most highly populated classes among all phages (International Committee on Taxonomy of Viruses), while other phages were found to interact with only one host. *Burkholderia multivorans* genome hosted seven different spacer sequences indicating the genus is more prone to bacteriophage infection. *Burkholderia* assembled genome interacted with four phages; three phages have interacted with the *Oxalobacteriaceae* genome, and one phage has interacted with the *Acinetobacter* genome. On analysis of CRISPRone, only in Maxbin 01 CRISPR-cas system was detected, and it is under the CLASS – 1, TYPE – 1, SUBTYPE—I – A. On other quality bins, the CAS system was not detected (Table 1).

Conclusion

This study performed a culture-independent approach for analysing the CRISPR arrays in MAGs to gain insights into phage interaction with bacterial genome. Studies on

Table 1 Phages interacted with the metagenome-assembled genomes

MGAs	CRISPR-Cas systems	CRISPR array	Spacer sequence	Phages	Life cycle
Burkholderia	Class-I, Type-I, Subtypes – I – A	Array_1	GAGCGCGCCACGGTCTGACGCCACCTGATCGCCGCCACGTACGACGCCACG CGCGCGCCACGCG	No hits	-
			CGGGCGCCACGCGCGCGCCGCCACGGACGACGCCACGGACGCCGCGC ACGGCGCGCCACGCGGTGACGCCACGGA		
			GTGCGCCACGACGACGCCACGCGCGCGCCGCCATGACGCGCGCC ACGGTTCGCGCCACGGT		
			TACGACGCCACGACGCCGCCACGCTACGACGCCACGGACGCCGCGCCACG GTGCGCGCCACGGG		
		Array_2	CGTGCCCGAGGTACCGGAAGTGC CGGACGTG	No hits	
		Array_3	CGTTATGCGCAACAGCCACGCAAACTGCGATTTCACACCATGG TGGAAAGGGA	Uncultured Caudovirales Phage (Caudoviricetes – Order) (Accession—LR796167.1)	Lytic or Lysogenic (based on the host environment)
			GACGACACATATACAAACTT		
		Array_4	CGCCCGAAGTGCCGCCACCTGAGGTGCTCCACCGAAGTGCCGCCAC CTGAGGTG	Inoviridae sp.isolate ctuo07 (Tubulavirales – order) (Inoviridae -Family) (Accession—BK034467.1)	Lysogenic cycle
		Array_5	GTAACCGCTGAAGCGCTAACTCGGGT		
		Array_6	TACCGTACCGACTACACCACCAAGTCCCGCGGTGCGGATCG		
Array_7	GCAATCCGGCAGGCGGTGACCGGT				
Array_8	TGTGGCGCTGCTGAGGGCTTCGGCTTCGGC GGTCCGCTCGCGGTGGCGT	Brevundimonas phage vB_Bpos-Marchewka (Caudoviricetes – Order) (Accession—ON529851.1)	Lysogenic cycle		
Array_9	GGCGCCCGTGACTGCGCCGAGAACCCGGTGACGGCGCCACAGGATT TCCGCTGCCT	No hits			
Array_10	CGATGCCGGTCTGACGTCGAGGTGGACAGCGACAGGATCGTGTCTGTC	Burkholderia Phage Carl1 (Caudoviricetes – Order) (Accession—LR796167.1)	Lysogenic cycle		
Acinetobacter	No cas genes	Array_11	CGAAGTCGAGAGCGACGTGATCGAGC	No hits	-
		Array_1	TAAGATTTAAACAAATGGCTCAAAATGAGGGGATTCCCTACCAAACGCTGA TTGCTTCAGTTCGCAAGGT TTTAGCGCTTCTGCTGCTTTTGTCT	Caudoviricetes sp. isolate criUp1 (Caudoviricetes -order) (Accession – BK042771.1)	Lytic or Lysogenic cycle (based on the host environment)
Oxalobacteriaceae	No cas genes	Array_1	CGTGTGCACTAAGCCACAATAAAAAATGGATCGCGCGACGGGTTAGCAG CGT	Caudoviricetes sp.isolate crvpyl1 (Caudoviricetes – Order) (Accession—BK052428.1)	Lytic or Lysogenic cycle (based on the host environment)
		Array_2	AAATACATCTTGATATTAACCCCGCATTTTAAAGCATAATAGTAGTATTGTTA TTCAGA	uncultured Caudovirales phage (Caudoviricetes – Order) (Accession—LR796167.1)	Lytic or Lysogenic cycle (based on the host environment)
		Array_3	TACGCTCGCGACGGTTCGCCCTTCGCGCGGTTTTCGTTCCGCGTCAC GCGGCTC	Caudoviricetes sp. isolate ctP2k10 (Caudoviricetes – Order) (Accession – BK018524.1)	Lytic or Lysogenic cycle (based on the host environment)
		Array_4	GGTCCGAAGTCGAGCCTCTAGAGTCGAAGTCGAGCTTCCGAGGGTCTGA AGTCGAGCCTCGG	No hits	-

Table 1 (continued)

MAGs	CRISPR-Cas systems	CRISPR array	Spacer sequence	Phages	Life cycle
<i>Burkholderia multivorans</i>	No cas genes	Array_1	GGGGTCCACGACCGCGAGTCGGCC	No hits	Lysogenic cycle
		Array_2	AGTGCGCCGGTGACGGTCTGAGACGAGCGCGGTGACGGGG	Ralstonia Phage Simangalove (Caudoviricetes – Order) (Bakolyvirus – Family) (Accession – NC_054946.1)	
		Array_3	CGAGCTGGTGGCGGACGAGAGCGCGGACGAGGTCTGAGGTCTGACAGCCGAAGCGAGCGAGCTATCGGTCTGA	Brevundimonas phage vB_Bpos Kabachok (Caudoviricetes – Order) (Accession – ON529852.1)	Lysogenic cycle
		Array_4	GCCGGTCTGACAGAGCCGATCTGAGTCTGTGGCTGACGAGAGGCCCGGTCTGAAGTCTGAG	No hits	-
			CCCGGTCTGAGAGAACCGATCTGAGCTGTAGCCGAGAGAGAGGCCCGGTCTGAGTCTGAA		
			GCCGGTCTGAAAGTCTGAGGTCTGAGAGCGAGGCGACTGAGCTATCGGTCTGA		
			GCTGAGGCCG		
			GCTGGTCTGAAAGTCTGAGGTCTGACAGCGAGCGAGCGACTGTGTGGTCTGCTGGTCTGA		
			GCTGAGGCCG		
			CGGAGCTGGACAGCGAGCTAACGGAGCTGTGTGGTCTGAGCTGAGACCCGTCGACACGACCCGATG		
			CCGAGCTGGACAGCGAGCTAACGGAGCTATTGGTCTGAGCTAAGACCCGTCGACAGCGATCTCGATC		
			ACCAGCACCGACACCGACAGCACCGACCGACCGACCGACCGACCGACCGT		
		Array_5	TATCTGTGACATCTCAAAATGCGGACCTCTATAGCACATATAAACGCGACAGACAAGCAGCAATACTAAAAATGGA	Chrysochromulina parva vibrophage larry (Priklausovirales – Order) (Lavidaviridae – Family) (Accession – MH920636.1)	Lytic cycle
		Array_6	TAATGCAATTGCAATGTTTATCTGCTCCAAACCCCAAGAAATAGTTTCAAAATCGTTCCGTAAAAAGGAACAACCTCAGGT	Uncultured Caudovirales phage (Caudoviricetes – Order) (Accession – LR796167.1)	Lytic or Lysogenic (based on the host environment)
		Array_7	GGGCCCCCATGCAAAAGCTGTCTGATCCG	No hits	-
			TGACTCTCATGCAAAAGCTGTGGCCCG		
			CATGCTTCGGGTGGGCTCGGTAG		
			GCAGCACCGCGCAAGAGAGCGCGCTGCGAAGAGGTCTGCA	Gordonia phage starstruck (Caudoviricetes – Order) (Emilbongvirinae – Family) (Accession – ON456333.1)	Lytic cycle
			GTAGCGGCGAAGAGGCGCACCGGCGAAGAGGCTGTCTGCGAAGAAGGCGGACCCG	Gordonia phage Bakery (Caudoviricetes – order) (Accession – NC_055804.1)	
			AAGGCCGCGCGCGAAGAGGCTGTCTGCGC	Gordonia phage Pons (Caudoviricetes – order) (Accession – NC_070891)	

phage interaction in MAGs can pave a way to understanding the phage population interactive with uncultivable bacteria. This can lead better understanding of defence mechanism of uncultivable bacteria against their bacteriophage. However, tools and algorithms available for the construction of MAGs and detection of CRISPR elements are far less limited. Specific CRISPR detection tools need to be developed for metagenomic analysis.

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Author contributions AA: conceptualization; data curation; formal analysis; methodology; validation; visualization; writing—original draft; writing—review and editing.

LM: conceptualization; investigation; project administration; resources; software; supervision; validation.

PK: writing—review and editing.

Data availability The data presented in the study are deposited in the NCBI SRA repository with the link <http://www.ncbi.nlm.nih.gov/bioproject/941822> and accession number—PRJNA941822.

Declarations

Conflict of interest The authors declare no competing interests.

Competing interests The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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